Leveraging Interpolation Models and Error Bounds for Verifiable Scientific Machine Learning

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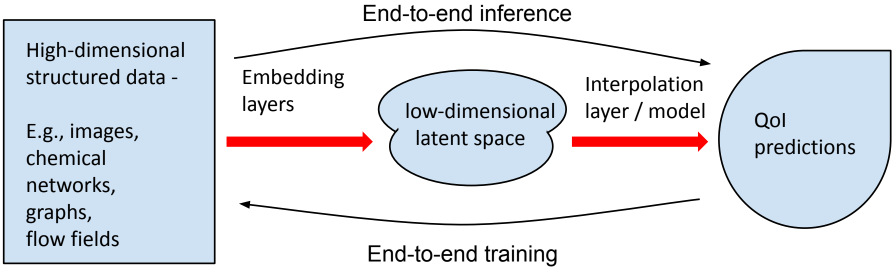
AI, SciML, and Interpolation

AI $=$ *Representation Learning* $+$ *Regression/Interpolation*



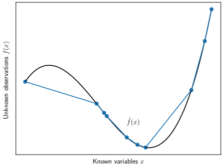
SciML $=$ **separate** representation learning and regression or interpolation

This paper $=$ **just interpolation** in the context of SciML

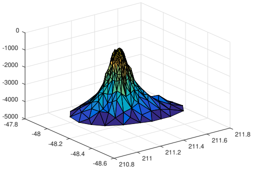
Outlines

Piecewise Linear (Delaunay) Interpolation

Approximate $f(q)$ by piecewise linear $\hat{f}_T(q)$; T = simplicial mesh



1 dimension



d dimensions

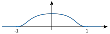
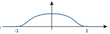
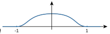
If $S \in T$ contain q ; diameter$(S) = h$:

$$|f(q) - \hat{f}_T(q)| \leq \left(\frac{\gamma + \gamma\sigma\sqrt{d}}{2} \right) h^2$$

γ = Lip const of ∇f; σ = condition number of interpolation matrix (skew of S);

RBF Interpolation (thin-plate spline kernel)

Approximate $f(q)$ by $\hat{f}_{RBF}(q)$ = linear combination of radial basis functions (RBFs)

$\hat{f}(q) = w_1$  $+ w_2$  $+ \ldots + w_n$  $+ \, tail(\mathrm{q})$

We use RBF = thin-plate-spline (TPS) in this work – it is the default in SciPy

If h = distance from q to the nearest interpolation node (training point), then:

$$|f(q) - \hat{f}_{RBF}(q)| \leq C \cdot h\sqrt{\log(h^{-1})}$$

C is a constant depends on f, the choice of RBF, and (indirectly) on d, but independent of h

Gaussian process interpolation

Two interpretations:

1. $\hat{f}_{GP}$ = RBF interpolant with "Gaussian bump" kernel
2. $\hat{f}_{GP}$ = posterior distribution over all possible realizations of f given the observations in the training data

(2) lets us use the variance of q as an uncertainty heuristic:

$$K(q,q) = \tau_f \left(1 - \|\phi^{(1)}(q), \ldots, \phi^{(1)}(q)\|^2_{A_{GP}^{-1}}\right)$$

$\phi^{(1)}$, …, $\phi^{(n)}$ are Gaussian basis functions; $\tau_f \sim$ variance of f; $\|\cdot\|_{A_{GP}^{-1}}$ is a rescaling of $\|\cdot\|_2$

Neural Network models (MLPs)

In a typical modern AI model:

$$\hat{f}_{ANN}(Q) = \hat{f}_{MLP} \circ \hat{f}_R(Q)$$

where

$$\hat{f}_R(Q) : ? \to \mathbb{R}^d$$

and

$$\hat{f}_{MLP}(q) = \ell^{(N)} \circ \ell^{(N-1)} \circ \ldots \circ \ell^{(0)}(q)$$

Each

$$\ell^{(i)}(x^{(i)}) = u\left(W^{(i)} x^{(i)} + b^{(i)}\right)$$

where $W^{(i)}$ and $b^{(i)}$ define a linear transformation, and u is a nonlinear "activation function"

We will focus on MLPs which use ReLU (piecewise linear) activation function which is the most common option

Experiments on synthetic datasets

For all the interpolation models we saw constants that depend on:

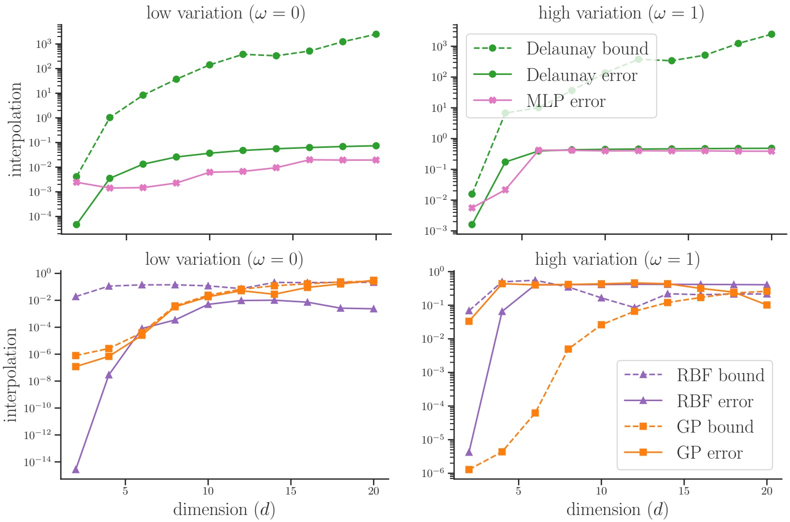
1. the dimension of the problem
2. the density of the training data,
3. the smoothness of f,
4. the conditioning of the interpolation matrices,

We generate hundreds of datasets from different distributions varying the

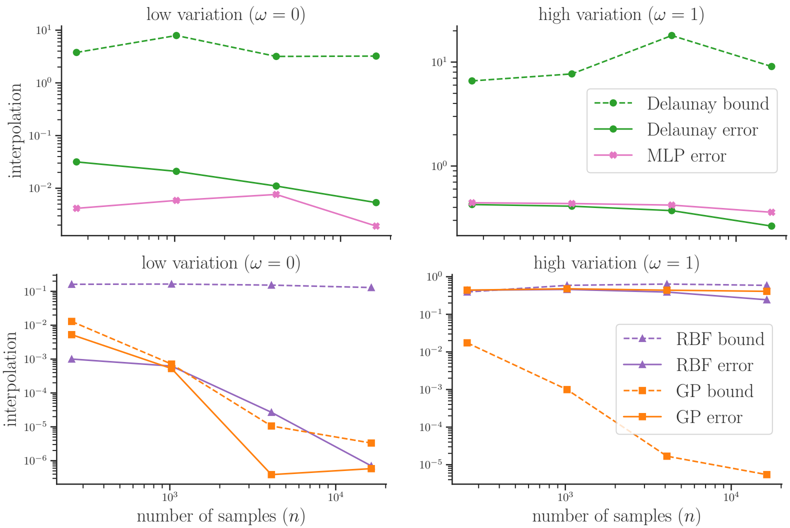
1. dimension d
2. number of training points n
3. Lipschitz constant of the underlying f
4. skewness of the training points

We use all methods on all training sets and measure the prediction errors on a holdout set generated from the same distribution.

Effects of dimension and smoothness



Effects of training sample size and smoothness



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Questions

Introduction and SciML Context

Interpolation Models and Error Bounds

Experiments on Synthetic Data

Experiments on Airfoil SciML Problem

github.com/LLNL/interpML

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